

Operators manual for Attachment



Manufacturer:

AVANT[®]
AVANT TECNO OY
e-mail: sales@avanttecno.com

Ylötie 1
FIN-33470 YLÖJÄRVI
FINLAND
Tel. +358 3 347 8800
Fax +358 3 348 5511

www.avanttecno.com

1. Introduction

Avant Tecno Oy wants to thank you for purchasing this original Avant attachment. It is the result of Avant's long experience in design and manufacturing of attachments. We ask you that you read and understand the contents of this manual completely before operating the attachment. This will improve your operating and maintenance efficiency, help avoid breakdowns and damage and extend your machines life. Do not hesitate to contact your local Avant dealer for advice regarding service, parts and any problems relating to the operation of your attachment.

2. Designed purpose

The Avant 1200 mm lawn mower is a horizontal plane cutter and it is meant only for finishing the lawn. The mower is meant for short grass only, not for i.e. long hay. Cut the lawn often enough, at least once in a week, so that the cutting waste stays short enough and decomposes rapidly to the root level. A good rule would be to cut no more than 1/3 of the grass at a time. When the grass is too long the cutting waste tends to pile up as lumps in the lawn. If this happens you have to cut the grass one more time to spread the waste evenly on to the lawn.

3. Safety instructions



- Never use the mower without instructions. Read all the decals and this manual.
- Start the operation slowly and carefully.
- Do not wear loose clothing, long uncovered hair or jewellery near the machine.
- When coupling the attachment, make sure that the attachment locking pin locks in positively.
- Keep hands, feet and clothing away from any moving part (e.g. cutting blades or belt)
- Make sure, that there are no bystanders in the working area, because the mower can throw small stones etc. from the lawn.
- Never put any part of the body under the mower deck, while the engine is running.
- Always carry the mower as low as possible.
- Never perform any service operations when the engine is running.

4. Technical specification

Cutting width	1200 mm
Width	1300 mm
Length	1150 mm
Height	370 – 470 mm
Weight	160 kg
Number of blades	2 pcs
Cutting height	30 – 130 mm
Blades are hydraulically driven	

- The mower is equipped with vertical floating, which enables cutting on uneven surfaces.
- The cutting height of the mower can be adjusted with plastic rings around the axels of the wheel supports.
- For security reasons the mower is equipped with a valve, which stops the blades when the mower is lifted.

5. Attaching the mower deck to the loader

5.1 Mechanical attachment

Coupling the attachment to a 500 series loader happens as follows:

Stage 1:

- Lift up the two locking pins on either side of the attachment coupling plate and turn them backward so that they remain in the locked up position.
- Make sure, that the pins remain in the up position; otherwise you cannot couple the attachment properly!



Stage 2:

- Turn the attachment coupling plate with the tilt movement so that the upper edge of the plate leans forward.
- Drive the loader into the attachment.



Stage 3:

- Lift the boom a little so that the attachment lifts off of the ground.
- Pull the boom control lever to the left so that the lower edge of the attachment coupling plate turns into the attachment.



Stage 4:

- Pull the boom control lever more to the left so that attachment coupling plate turns more and the bolts on the boom push the pins down in the holes of the attachment.
- **CAUTION! Make sure that the pins lock properly down in the holes of the attachment.**



Always remember the safety instructions from paragraph 3, when attaching silage forks.

Coupling the hydraulic hoses to a 500 series loader

The mower deck has three hydraulic quick couplings, which should be attached to the corresponding couplings in the loader. There are two male couplings and one female coupling in the mower deck and correspondingly the loader has two female couplings and one male coupling.

The hoses from the mower deck should be attached to the loader correctly (pressure line to pressure line and return line to return line). The blue rubber caps of the couplings always mark the pressure line (one on the loader and one on the attachment). Always connect them as a pair. This way the mower deck will always start to rotate correctly when the auxiliary hydraulics lever is moved.



If the mower deck does not start to rotate when the loader's auxiliary hydraulics control lever moved, please change the coupling of the return- and pressure hose. **See picture 1**



Picture 1



Make sure that the hydraulic hoses **do not get overstretched** in any boom or attachment plate position and are **not in a position** where they can be **trapped during the operation** of the machine and attachment.

- For coupling the hydraulic hoses see also the operators manual of the loader.
- Always remember the safety instructions from paragraph 3, when coupling the attachment.

Coupling the attachment to a 200 series loader happens as follows:

Stage 1:

- Lift up the locking pin and turn it backward so that it remains in the locked up position.
- Make sure that the pin remains in the up position, otherwise you cannot couple the attachment properly!



Stage 2:

- Turn the attachment coupling plate with the tilt movement so that the upper edge of the plate leans forward.
- Drive the loader into the attachment so that the pins in the upper left and right corner of the attachment coupling plate go into the holes of the attachment bracket.



Stage 3:

- Lift the boom a little so that the attachment lifts off the ground.
- Pull the boom control lever to the left so that the lower edge of the attachment coupling plate turns into the attachment



Stage 4:

- Pull the boom control lever more to the left so that the attachment coupling plate turns more and the boom pushes the pin down in the hole of the attachment.
- **CAUTION! Make sure that the pin locks properly down in the hole of the attachment.**



Coupling the hydraulic hoses to a 200 series loader

The mower has three hydraulic quick couplings, which should be attached to the corresponding couplings in the loader. There are two male couplings and one female coupling in the mower. The loader has correspondingly one male coupling and two female couplings.

The hoses from the mower deck should be attached to the loader correctly (pressure line to pressure line and return line to return line). The red caps of the couplings always mark the pressure line (one on the loader and one on the attachment). Only this way the mower starts to rotate, when the loaders auxiliary hydraulics pedal is pressed down to its locked position.

The 200 series loaders and their attachments are equipped with flat face couplings, because they are more convenient and clean in everyday use. However there are some things you have to remember when coupling and uncoupling a flat face coupling.

1. Always try to **keep the couplings as clean as possible**. Dirt, ice etc might prevent the coupling from functioning properly.
2. Although the joint surface seems flat there are sprung parts in the coupling, which give in when pressed together correctly. If the couplings do not lock in to each other try **to release pressure by moving the auxiliary hydraulics pedal up and down** a couple of times and then try again.

3. **Before uncoupling** the couplings **always stop the engine** from of the loader and **move the auxiliary hydraulics pedal up and down** a couple of times. This releases any pressure left to the attachment and ensures easy coupling the next time you are using the attachment.
4. Flat face couplings have a locking mechanism, which prevents them to uncouple accidentally. When removing them from each other make sure that the ball and the slot from the female coupling are facing each other. **See picture 2**



Picture 2



Always **make sure**, that **all couplings are properly locked in before starting the engine**. The hydraulic motor of the mower may get damaged, if one of the couplings is not locked in.



If the female coupling will not release from the male coupling when its collar is pulled backwards, rotate the collar until the slot in it is facing the ball (**see picture 2**) and try again.



If the mower does not start to rotate when the loader's auxiliary hydraulics pedal is pressed down to its locked position, please change the coupling of the return- and pressure hose. **See picture 3**



Picture 3



Make sure that the hydraulic hoses **do not get overstretched** in any boom or attachment plate position and are **not in a position** where they can be **trapped during the operation** of the machine and attachment.

- For coupling the hydraulic hoses see also the operators manual of the loader.
- Always remember the safety instructions from paragraph 3, when coupling the attachments.

6. Instructions for use

6.1 Adjusting the cutting height

The cutting height of the mower can be adjusted with plastic rings around the axel of the wheel supports.

Adjusting is done as follows:

1. Lift the mower up with your loader, turn the engine off and put on the service support (see the users manual of the loader).
2. Open and remove the pin from the vertical axel of the wheel support. (Picture 4)



Picture 4

3. Pull the wheel support away from its bracket.
4. Move the plastic rings to the upper side of the bracket if you wish to lower the mower and other way round if you wish to increase cutting height. (See picture 5)



Picture 5

5. Push the vertical wheel support axel back through the bracket and put the pin back on.

6. Repeat phases 2 – 5 with all the other wheels as well. When you are finished, make sure, that there are equal amount of same sized plastic rings on top of each bracket.

6.2 Floating

The mower is equipped with a floating, which enables mowing on uneven surfaces. Floating is the mechanism, which allows the mower deck to go up and down according to the terrain, without moving the boom from the loader.

In order to achieve the best lawn mowing result available, the correct adjusting of the floating-device becomes essentially important. If the setting has been left too high, the cutter blades of the lawn mower remain hanging up in sudden descends. If it is left too low in steep raising grounds, the front wheels of the loader will be raised up, and the lawn mower intends to become a plough.



NOTE! If the floating is in its lowest position the blades will stop to rotate, because the safety valve is turned on.



If the blades are not rotating or rotating only from time to time, lift the loader boom higher.

Adjusting the floating



Picture 6

In **picture 6** the floating is correctly adjusted. The floating boom can move freely up and down according to terrain.



Picture 7

In **picture 7** the floating is adjusted too low. If the floating is in the lowest position it activates the safety valve and stops the blades.



Picture 8

In **picture 8** the floating is adjusted too high. The floating boom cannot move upwards and the loader's boom tends to lift the mower in to air in sudden downhill.

6.3 Side discharge for cutting waste

The mower deck is equipped with an optional side discharge. If you wish to use the mower deck as a side discharging model you should mount the side discharge tube (supplied with the mower) on to the mower. The mounting is done as follows:

1. Lower the mower on to the ground and turn off the loader's engine.
2. When viewed from front on the left side of the mower is a cover plate. Remove the three bolts holding the cover and remove the cover plate, also remove the two bolts on top of the deck. The bolts can be opened with a 13 mm spanner.
3. Mount the supplied side discharge tube on to the mower as shown in **picture 9** and tighten all the five bolts shown in **picture 9**.



Picture 9

7. Service

7.1 Greasing points

The mower deck has six greasing points, which need to be greased 1 - 3 times a year. Four of them are located in the vertical axel bracket of the wheels and two of them are at the lower end of the floating boom. (See **pictures 10** and **11**)



Picture 10



Picture 11



Do not over grease the greasing points. A couple of presses with the grease press is enough.

7.2 Tightening the cogged belt

The tightness of the cogged belt should be checked always at the beginning and end of a cutting season. The tightness is checked and adjusted as follows:

1. Remove the cover on top of the mower.
2. Remove hose cover above the belt pulley
3. Loose the four bolts underneath the belt pulley, in order to tighten the belt. (See **picture 12**)



Picture 12

4. The actual tightening is done by tightening two nuts on the left-hand-side of the mower. Tighten these nuts equally from both sides. (See **picture 13**)



Picture 13

5. The belt should not be tightened too much, but it should not stay too loose either. When the belt is pushed horizontally from the point the arrow is showing with power of 2 kg it should bend 5mm. (See **picture 14**)



Picture 14

6. When the belt is correctly tightened, tighten the four nuts under the belt pulley, which were loosened at **phase 3**.
7. Attach the covers, which were removed at **phase 2** and **1** and you have finished.

7.3 Changing and replacing the cutting blades

Check the condition of the blades after every two or three cutting times. If the quality of cutting has significantly reduced from original, it is probable, that the blades have gotten dull.

It is possible to sharpen the blades, but if there are deep dents or chips, it is reasonable to change the blades to new ones.

- The product numbers are:

The bigger lower blade	A33205
The smaller upper blade	A46612

7.4 Washing your lawn mower

- It is permitted to wash the mower with a power washer.
- Do not use any strong dissolvent when washing.

7.5 Protective bolts

The blades of the mower are equipped with protective bolts, which will break, if the blades hit a big stone or other heavy obstacle. This prevents any further damage to the mower i.e. belts, bearings and hydraulic motor should stay in working order. The breaking bolts are of size M6*25.

7.6 Synchronization of the blades

The mower has two blades, and their revolution orbits go partially on top of each other. It is important, that the synchronization is correctly set up, because otherwise the blades will collide.

The synchronization is correctly adjusted when it leaves the factory. But after adjusting the belt, or after the blades have hit an obstacle, it is recommended to check the synchronization as well.



If you hear a clinking sound when cutting, check the synchronization immediately.

- **Checking the synchronization**

1. Lift the mower with your loader, attach the service support (see loaders owners manual) and turn off the engine.
2. Rotate the blades by hand.
3. When one of the blades is vertical the other one should be horizontal.
4. If the blades collide when rotated by hand adjusting is necessary.

- **Adjusting the synchronization**

1. Remove the cover on top of the mower.
2. Remove hose cover above the belt pulley.
3. Loose the four bolts underneath the belt pulley, in order to loosen the belt. (See **picture 9**)
4. Loosen two screws pointed by **picture 10** so much, that you are able to rotate the belt pulley without moving the belt.
5. Now rotate the belt pulley from **picture 9** while holding the belt in place so much, that the blades are correctly adjusted.
6. When the blades are correctly synchronized tighten the cogged belt as advised above and attach all covers in place.

8. Warranty terms for Avant 1200 mm mower deck (Product number A 21046)

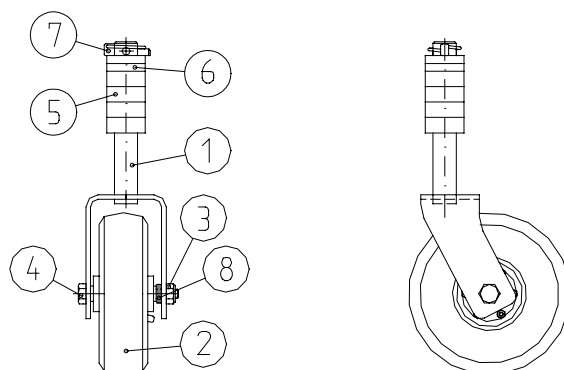
During the first year of operation Avant Tecno Oy warrants to replace any part or repair any defect, which may occur, subject to the terms below:

1. The product has received regular maintenance in accordance with schedules given by the manufacturer.
2. Any damage caused by operation in a negligent manner or exceeding the approved specifications detailed in this manual is excluded.
3. Avant Tecno Oy accepts no responsibility for interruption to working or any other consequential losses resulting from any failure of the product.
4. Only Avant Tecno Oy approved replacement or original quality parts shall be used during routine maintenance.
5. The Avant Warranty excludes any consumable parts (e.g. tyres, blades or belts) except where it can be clearly shown that these parts were defective on original supply.

In the event a fault occurs which is attributable to manufacturing or assembly defect you should arrange to return your attachment to your authorised dealer for repair. Travel and freight costs are excluded.

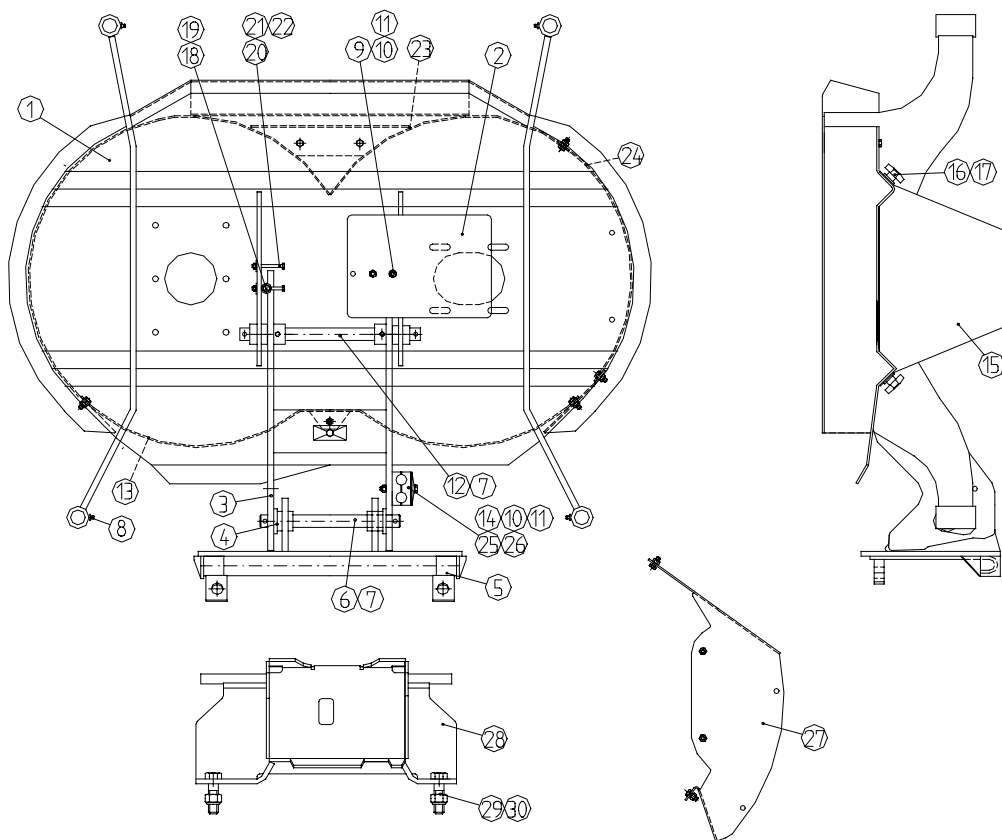
9. Mower deck (A21046) spare parts list

SUPPORT WHEEL A46576



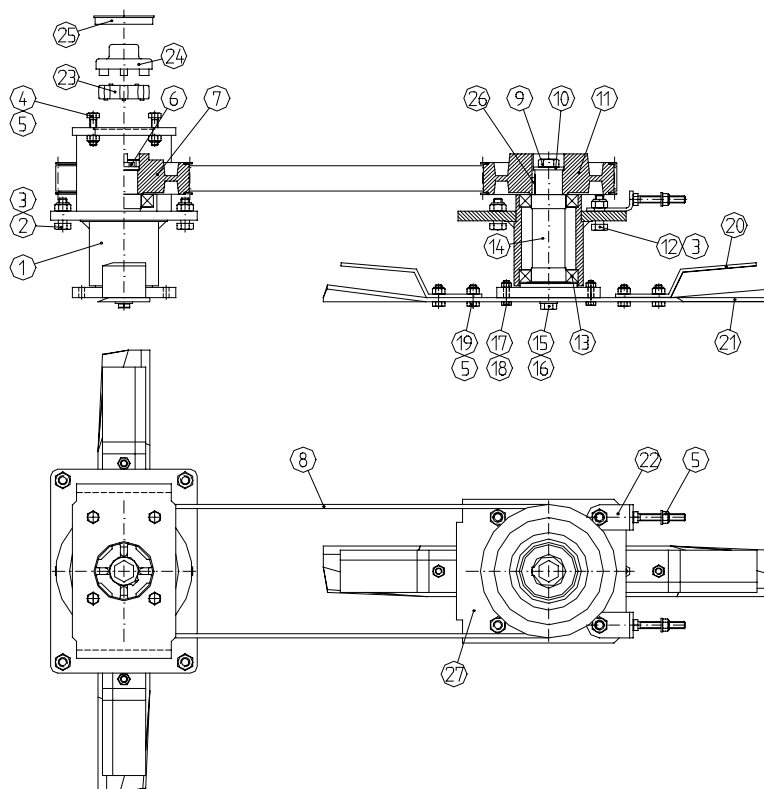
1	A46574	1	YOKE, SUPPORT WHEEL
2	A46561	1	WHEEL 250-3
3	72324	1	HEXAGONAL NUT M12 DIN 985
4	73573	1	HEXAGONAL SCREW M12*120 ZN DIN 933
5	A44665	4	ADJUSTING RING, PLASTIC 20 MM
6	A44664	2	ADJUSTING RING, PLASTIC 10 MM
7	71727	1	RING SPLINT 9,5 MM ZN
8	71691	2	BUSHING M12 ZN DIN 125

FRAME AND COVERS



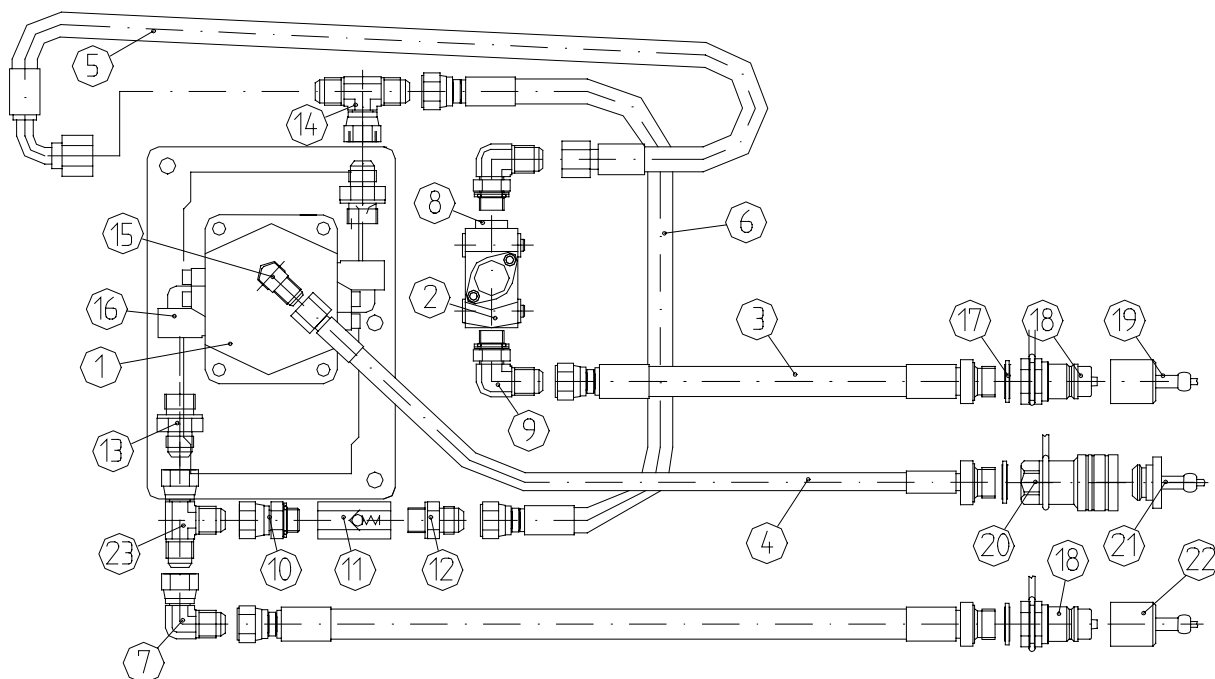
1	A21121	1	FRAME
2	A46547	1	HOSE PROTECTOR PLATE
3	A33206	1	FLOATING DEVICE
4	A46019	2	SLIDING SLEEVE
5	A33207	1	ATTACHMENT PLATE
6	A46566	1	PIVOT PIN, REAR
7	71774	4	SPRING SPLINT 8*40 DIN 1481
8	72798	6	GREASE NIPPEL R1/8 STRAIGHT DIN 71412
9	72132	2	HEXAGONAL SCREW M8*25 DIN 933 ZN
10	72320	4	HEXAGONAL NUT M8 DIN 985
11	73122	4	BUSHING M8 DIN 125 ZN
12	A46567	1	PIVOT PIN, FRONT
13	A33210	1	MULCHING PLATE, REAR
14	72141	2	HEXAGONAL SCREW M8*70 DIN 931 ZN
15	A33211	1	COVER
16	64355	4	FINGER SCREW M8
17	71685	4	WASHER, LARGE M8 DIN 440 ZN
18	74138	1	HEXAGONAL SCREW M8*55 DIN 933 ZN
19	72319	1	HEXAGONAL NUT M8 DIN 934 ZN
20	72125	3	HEXAGONAL SCREW M6*60 DIN931 ZN
21	72318	3	HEXAGONAL NUT M6 DIN 985
22	71681	3	WASHER, LARGE M6 DIN 440 ZN
23	A46823	1	MULCHING PLATE, FRONT
24	A46824	1	SIDE DISCHARGE COVER PLATE
25	64890	2	COVER PLATE FOR HOSE CLAMP F22
26	64889	2	HOSE CLAMP F22
27	A46826	1	SIDE DISCHARGE TUBE KP.
28	A33126	1	ADAPTER, 200-SER.
29	72229	2	HEXAGONAL SCREW M20*70 DIN 931 ZN
30	72335	2	HEXAGONAL NUT M20 DIN 985

TRANSMISSION



1	A46592	1	TUBE, LEFT
2	72151	8	HEXAGONAL SCREW M10*30 DIN 933 ZN
3	72322	10	HEXAGONAL NUT M10 DIN 985 NYLOC
4	73443	4	HEXAGONAL SCREW M8*35 DIN 933 ZN
5	72320	14	HEXAGONAL NUT M8 DIN 985 NYLOC
6	A46608	1	SPECIAL BOLT
7	A46597	1	COGGED BELT PULLEY, LEFT
8	64877	1	COGGED BELT
9	72183	1	HEXAGONAL SCREW M14*25 DIN 933 ZN
10	A46595	2	BUSHING
11	A46598	1	COGGED BELT PULLEY, RIGHT
12	74505	2	HEXAGONAL SCREW M10*35 DIN 933 ZN
13	64038	4	BEARING 6008 2RS
14	A33099	2	AXLE
15	73041	2	HEXAGONAL SCREW M12*25 DIN 933 ZN
16	71691	2	BUSHING M12 DIN 125 ZN
17	72119	4	HEXAGONAL SCREW M6*25 DIN 933 ZN
18	72318	4	HEXAGONAL NUT M6 DIN 985 NYLOC
19	72130	8	HEXAGONAL SCREW M8*20 DIN 933 ZN
20	A46612	4	MULCHING BLADE
21	A33205	2	BLADE
22	A46828	2	COGGED BELT TIGHTENER
23	64299	1	RUBBER FOR CLUTCH
24	64301	1	CLUTCH HALF
25	A46596	1	FITTING RING
26	A46582	2	WEDGE
27	A46564	1	TUBE, RIGHT

HYDRAULIC HOSES



1	64876	1	HYDRAULIC MOTOR PLM 20.11,2
2	64761	1	STOPPER VALVE DF5/3A 17C-BSP3/8
3	A47267	2	HYDRAULIC HOSE L=2380 2LW-06
4	A47268	1	HYDRAULIC HOSE L=2380 1LW-04
5	A44336	1	HYDRAULIC HOSE L=360 2LW-06
6	A44024	1	HYDRAULIC HOSE L=340 2LW-06
7	64134	1	ELBOW FITTING UNF3/4 SK-UNF3/4 UK
8	74411	1	PLUG R3/8
9	64068	2	ELBOW FITTING R3/8 UK-UNF3/4 UK
10	64082	1	CONNECTOR FITTING R3/8 UK-UNF3/4 SK
11	64065	1	VALVE VUC 20
12	64042	1	FITTING R3/8 UK-UNF3/4 UK
13	64058	2	FITTING R1/2 UK-UNF3/4 UK
14	64307	1	T-FITTING UNF3/4 SK-2*UNF3/4 UK
15	64087	1	ELBOW FITTING R1/4 UK-UNF9/16 UK
16	64067	2	PRESSURE FLANGE 1G12
17	73047	3	USIT-FITTING R1/2
18	64563	2	QUICK COUPLING R1/2 SK FLAT, 200-SER.
18	73409	2	QUICK COUPLING R1/2 SK, 500-SER.
19	64811	1	COVER FOR QUICK COUPLING, RED, 200-SER.
19	64878	1	COVER FOR QUICK COUPLING, BLUE, 500-SER.
20	64564	1	QUICK COUPLING R1/2 SK FLAT, 200-SER.
20	73457	1	QUICK COUPLING R1/2 SK, 500-SER.
21	64568	1	COVER FOR QUICK COUPLING, BLACK, 200-SER.
21	73754	1	COVER FOR QUICK COUPLING, BLACK, 500-SER.
22	64567	1	COVER FOR QUICK COUPLING, BLACK, 200-SER.
22	73753	1	COVER FOR QUICK COUPLING, BLACK, 500-SER.
23	64050	1	L-FITTING UNF3/4 SK-2*UNF3/4 UK
24	A33107	1	PROTECTIVE SPIRAL SP-032 O.5M